DIVIDING A MODULE BY ITS TORSION IN THE IDEAL MAKES THE LOCAL COHOMOLOGY CONCENTRATE IN DEGREE ZERO

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The identification of the zeroth local cohomology with support in a finitely generated ideal with the kernel of the map to the localizations. Invoked at the start of the counted proof (source lines 2127-2131). Computes the zeroth cohomology with support and proves the equality with the closed subscheme version.
- The vanishing of the higher local cohomology of a torsion module. Invoked inside the counted proof (source lines 2135-2136). Gives the vanishing of the higher cohomology of the torsion submodule.
- The characterization of modules without torsion as those annihilated by the functor from the proof of the first statement. Invoked inside the counted proof (source lines 2139-2140). Shows that the quotient by the torsion has no zeroth local cohomology.

Source reference. Original lemma lemma-adjoint: sources/stacks-dualizing/pinned-original-source.tex, source lines 1294–1294. Navigation pointer only: the counted text refers to the Stacks item lemma-adjoint, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Source reference. Original lemma lemma-local-cohomology-adjoint: sources/stacks-dualizing/pinned-original-source.tex, source lines 1466–1466. Navigation pointer only: the counted text refers to the Stacks item lemma-local-cohomology-adjoint, which is an existing labeled result of this same chapter and is not part of this card.

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Source reference. Original reference more-algebra-lemma-divide-by-torsion: sources/stacks-dualizing/pinned-original-source.tex, source lines 2140–2140. Navigation pointer only: the counted text refers to the Stacks item more-algebra-lemma-divide-by-torsion, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Lemma 0.1. *Let A be a ring and let $I \subset A$ be a finitely generated ideal. Let M be an A-module. Let $Z = V(I)$. Then $H^0_I(M) = H^0_Z(M)$. Let N be the common value and set $M' = M/N$. Then*

(1) *$H^0_I(M') = 0$ and $H^p_I(M) = H^p_I(M')$ and $H^p_I(N) = 0$ for all $p > 0$,*
(2) *$H^0_Z(M') = 0$ and $H^p_Z(M) = H^p_Z(M')$ and $H^p_Z(N) = 0$ for all $p > 0$.*

Proof. By definition $H^0_I(M) = M[I^\infty]$ is I-power torsion. By Lemma lemma-local-cohomology-adjoint we see that

$$H^0_Z(M) = \mathrm{Ker}(M \longrightarrow M_{f_1} \times \ldots \times M_{f_r})$$

if $I = (f_1, \ldots, f_r)$. Thus $H^0_I(M) \subset H^0_Z(M)$ and conversely, if $x \in H^0_Z(M)$, then it is annihilated by a $f_i^{e_i}$ for some $e_i \geq 1$ hence annihilated by some power of I. This proves the first equality and moreover N is I-power torsion. By Lemma lemma-adjoint we see that $R\Gamma_I(N) = N$. By Lemma lemma-local-cohomology-adjoint we see that $R\Gamma_Z(N) = N$. This proves the higher vanishing of $H^p_I(N)$ and $H^p_Z(N)$ in (1) and (2). The vanishing of $H^0_I(M')$ and $H^0_Z(M')$ follow from the preceding remarks and the fact that M' is I-power torsion free by More on Algebra, Lemma more-algebra-lemma-divide-by-torsion. The equality of higher cohomologies for M and M' follow immediately from the long exact cohomology sequence. □